10 **(a)** **(i)** State what is meant by a standard candle.

....

.... [1]

(ii) Explain how standard candles may be used to determine the distance of a galaxy from an observer.

....

....

.... [2]

(b) A star in a distant galaxy has luminosity L.

The star emits radiation that has maximum intensity at a wavelength that is known to be λ_{star} after correcting for redshift.

The Sun has surface temperature T_0 and emits radiation that has maximum intensity at wavelength λ_0.

(i) Explain why the wavelength of the radiation observed from the star has to be corrected for redshift.

....

....

.... [2]

(ii) State an expression, in terms of λ_{star}, λ_0 and T_0, for the surface temperature T_{star} of the star.

T_{star} = ... [1]

(iii) Determine an expression, in terms of L and T_{star}, for the radius of the star. Identify any other symbols that you use.

radius = ... [1]

[Total: 7]